PAPER_519 — Shell Radiance Prototype Equation: Full 26D Layer Formulation

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

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PAPER_519 — Shell Radiance Prototype Equation: Full 26D Layer Formulation

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§1 — Abstract

This paper assembles the complete Session 140 recalculation into a prototype shell radiance equation system, integrating: the upgraded time adjustment, dual existence mathematics, DPM-based ProtoH shell formation, universal buoyancy via layered radiance, BigBang trigger conditions, and updated probability of ordered structure with dilation-negative time factor.

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§2 — Upgraded Time Adjustment (Session 140 Canonical)

$$\boxed{t_{\text{adj}} = \frac{t_{\text{obs}}}{1 + \Delta_{\text{dil}}} + t_{\text{neg}}}$$

This supersedes the prior form $t_{\text{adj}} = t_{\text{obs}}/(1+\Delta_{\text{rel}})$ used in Session 136 which omitted t_{neg}. The upgrade restores the full bidirectional causality of the 26D shell system.

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§3 — Proto-Hydrogen 26D Layered Shells

$$\boxed{ProtoH = \emptyset^{26\,\text{layered shells}}

- \int DPM_{\text{react}}\, dt_{\text{adj}}

- Higgs_{\text{shift}} \cdot \sum_{\text{flavors}} RadianceEnergies
- DualExist_{\text{math}}}$$

Where:

- $\emptyset^{26}$: empty 26D shell alignments awaiting radiance
- $\int DPM_{\text{react}}\, dt_{\text{adj}}$: DPM filling over upgraded time
- $Higgs_{\text{shift}} \cdot \sum RadianceEnergies$: Higgs mechanism anchors

radiance energies to Standard Model mass flavors (6 quark + 6 lepton)

- $DualExist_{\text{math}}$: dual existence contribution

$= \int_{t_{\text{pos}}}^{t_{\text{neg}}} Existence\, dt$

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§4 — Universal Buoyancy via Shell Radiance

$$\boxed{U_b = F_{\text{inert}} \cdot Prob_{\text{order}}

- \frac{DPM_{\text{react}}}{UA_{\text{trapped}}}
- Higgs_{\text{shift}}
- \sum_{l=1}^{26} ShellEnergy^{(l)}}$$

This upgrades the prior U_b which lacked the $\sum ShellEnergy$ term — that term represents the direct radiance-buoyancy coupling absent in pre-Session-140 formulations.

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§5 — BigBang Initiation (DPM Reaction Trigger)

$$\boxed{BigBang = SCm_{\text{inj}} \cdot UA_{\text{contact}} \cdot DPM_{\text{react}} \cdot \sum_{s=1}^{N_{\text{clusters}}} Smalls^{26D\,\text{layered}} \cdot \exp(Grind_{\text{opp}})}$$

With t_{adj} upgraded, the trapped smalls are now calculated with: $$t_{\text{adj}} = t_{\text{obs}} / (1 + \Delta_{\text{dil}}) + t_{\text{neg}}$$

Opposite-side existence: $Existence_{\text{opp}} = DualExist(Existence_{\text{one}}, t_{\text{neg}})$

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§6 — Updated Probability of Ordered Structure

$$\boxed{Prob_{\text{order}} = \frac{\exp\!\left(-S_{26D\,\text{Egg}} / v_{\text{init}}\right)}{Partition_{9D}} \cdot (v_{\text{init}} - v_{\text{current}}) \cdot (1 + \Delta_{\text{dil}} \cdot t_{\text{neg}})}$$

Change from Session 136: the factor $(1 + \Delta_{\text{dil}} \cdot t_{\text{neg}})$ is new. Setting this factor to 1 (i.e., $\Delta_{\text{dil}} \cdot t_{\text{neg}} = 0$) recovers the Session 136 form — full backward compatibility.

Physical meaning: As dilation increases ($\Delta_{\text{dil}} \uparrow$) and $t_{\text{neg}} < 0$, the probability of ordered structure decreases, matching observed cosmic entropy increase.

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§7 — Shell Radiance Prototype: Full Assembly

The complete prototype shell radiance system for a 26D layered universe:

Equation	Role		
$ShellEnergy^{(l)} = \int Radiance_{\text{quant}} dt_{\text{neg}}$	Layer energy		
$t_{\text{adj}} = t_{\text{obs}}/(1+\Delta_{\text{dil}}) + t_{\text{neg}}$	Upgraded time		
$Distance_{\text{spooky}} = c \cdot	t_{\text{neg}}	$	Non-local inference range
$DualExist = \int_{t_{\text{pos}}}^{t_{\text{neg}}} Existence\, dt$	Bidirectional causality		
$F_{\text{inert}} = -\partial(DPM_{\text{react}} \cdot SE)/\partial v^{26} \cdot t_{\text{neg}}$	Inertial force		
$F_{\text{centrip}} = DPM_n \cdot \omega_{CW}^2 \cdot r^l / (1+\Delta_{\text{dil}})$	Shell coherence		
$F_{\text{centrif}} = DPM_s \cdot \omega_{CCW}^2 \cdot r^l \cdot t_{\text{neg}}$	BigBang expansion		
$Prob_{\text{order}} = \ldots \cdot (1 + \Delta_{\text{dil}} \cdot t_{\text{neg}})$	Structure probability		

Equation	Role		
$ProtoH = \emptyset^{26} + \int DPM_{\text{react}} dt_{\text{adj}} + \ldots$	Hydrogen genesis		
$U_b = F_{\text{inert}} \cdot Prob_{\text{order}} + \ldots + \sum SE$	Buoyancy		
$BigBang = SCm_{\text{inj}} \cdot UA_{\text{contact}} \cdot DPM_{\text{react}} \cdot \sum Smalls \cdot \exp(Grind)$	Trigger		

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§8 — Prototype Shell Radiance Equation (Master Form)

Combining §3–§6 into a single master expression for the 26D radiance state:

$$\Psi_{26D}(t_{\text{adj}}) = ProtoH + U_b \cdot Prob_{\text{order}} + BigBang \cdot \exp\!\left(-\frac{|t_{\text{neg}}|}{t_{\text{adj}}}\right)$$

This master form encodes the full evolution from empty shell alignments ($ProtoH$) through buoyant radiance ordering ($U_b \cdot Prob_{\text{order}}$) to the Big Bang trigger exponential decay as $|t_{\text{neg}}| \to t_{\text{adj}}$.

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§9 — Conclusion

The prototype shell radiance equation assembles all Session 140 upgrades into a coherent and internally consistent framework. The upgraded t_{adj}, updated $Prob_{\text{order}}$, and the full DPM force triad together resolve the mass-origin problem and unify observable cosmology with 26D UQFF geometry.

See also: PAPER_516 (DPM Shell Radiance), PAPER_517 (Negative Time Proof), PAPER_518 (DPM Forces), PAPER_520 (Session 140 Hub).